

- 2 (a) In the mammalian circulatory system, red blood cells travel through different types of blood vessel as they pass from the heart to respiring tissues and back to the heart.

Fig. 2.1 shows the types of blood vessels through which red blood cells travel in the circulatory system.

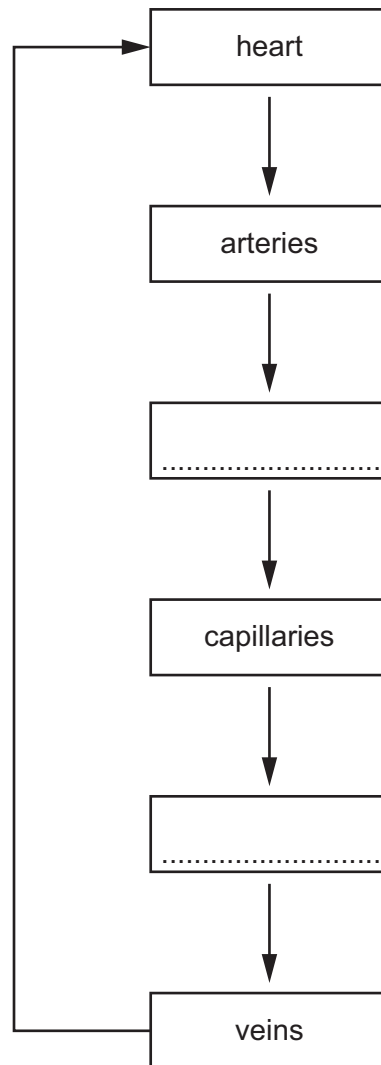


Fig. 2.1

Complete Fig. 2.1 by writing the names of the missing types of blood vessels through which red blood cells travel. [2]

- (b) Water is the main component of blood. It has an important role in the transport of substances around the body.

Fig. 2.2 shows the ionic compound sodium chloride dissolving in water.

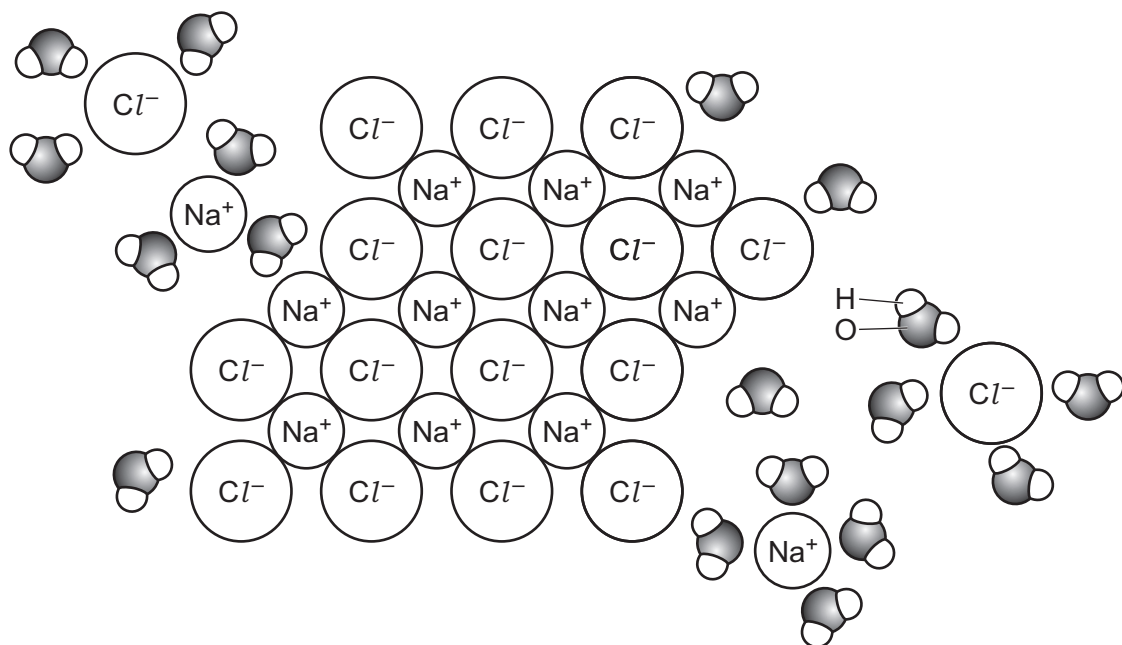


diagram not to scale

Fig. 2.2

With reference to Fig. 2.2, explain how water acts as a solvent for sodium chloride.

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(c) Fig. 2.3 shows a Galapagos penguin, *Spheniscus mendiculus*, swimming in the water.



Fig. 2.3

Penguins are birds that live on land but spend a lot of time swimming underwater hunting for food. Penguins can remain underwater for up to twenty minutes. During this time they do not breathe but their tissues continue to respire.

Haemoglobin in the red blood cells of penguins has a higher affinity for oxygen than haemoglobin in other birds that do not swim underwater.

Fig. 2.4 shows the oxygen dissociation curve for a bird that does **not** swim underwater.

- (i) Draw a line on Fig. 2.4 to suggest the position of the oxygen dissociation curve for penguin haemoglobin. [2]

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- (ii) Penguin haemoglobin is very sensitive to a decrease in pH caused by an increase in the carbon dioxide concentration in the blood.

Explain how a decrease in pH affects penguin haemoglobin, **and** suggest how this helps the penguin to swim underwater for a long time.

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- (d) The heart rate of a penguin decreases while it is swimming underwater.

Heart rate is regulated by a group of specialised cells in the wall of the right atrium. The activity of these cells is modified by nerve impulses.

Name the group of specialised cells in the wall of the right atrium that regulates heart rate.

..... [1]

[Total: 11]